

Terminal

Infrastructure: Power Grid

Budget: \$5,00,000

Location: Dubai, UAE

Target Year: 2040

Risk Level: LOW

Risk Score: 0.56/100

Climate Resilience Report for Dubai Power Grid

Given the coordinates [25.6649, 55.8230], [25.6640, 55.8244], [25.6629, 55.8231], [25.6642, 55.8219] in Dubai, and considering the recent severe flooding in April 2024 and the projected 0.7m sea level rise by 2100, this report outlines site-specific resilience strategies for the power grid infrastructure.

STRATEGY 1

Elevating Critical Infrastructure Components: To mitigate the risk of flood damage, critical power grid components such as substations and transformers can be elevated above the projected flood levels. This can be achieved through the construction of elevated platforms or by relocating these components to higher ground. By doing so, the power grid can continue to operate even during flooding events, ensuring a reliable supply of electricity to the city.

STRATEGY 2

Implementing Flood-Resistant Materials and Designs: Utilizing flood-resistant materials and designs for power grid infrastructure can enhance its resilience to flooding. This includes using materials such as stainless steel, fiberglass, or concrete for equipment enclosures and foundations, which can withstand floodwaters. Additionally, designing equipment with waterproof seals and protective coatings can prevent water ingress and minimize damage.

STRATEGY 3

Integrating Smart Grid Technologies and Flood Monitoring Systems: Implementing smart grid technologies and integrating flood monitoring systems can enable real-time monitoring of flood events and proactive management of the power grid. This includes installing sensors to detect water levels, water quality, and other parameters, which can provide early warnings of potential flooding. By leveraging these technologies, grid operators can take preventive measures such as switching to backup power sources or isolating affected areas to minimize disruptions.

RESILIENCE SCORE: 87%

